

Anticipating demand, not only carbon: receding-horizon scheduling of deferrable computing workloads across data centres with calibrated carbon-intensity forecasts

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Abstract

Shifting deferrable computing jobs to cleaner hours and locations cuts data-centre emissions, but online schedulers must act before future grid conditions and jobs are known. This paper formulates multi-site, deadline-constrained scheduling of bag-of-tasks jobs, including data-transfer emissions, as a minimum-cost flow problem. This yields exact offline optima and fast online planning, and it is proved that myopic receding-horizon scheduling can be arbitrarily worse than the offline optimum even with perfect carbon forecasts. The paper then proposes CARMA (Carbon-aware Anticipatory Receding-horizon Multi-site Allocation), which adds “ghost jobs” for expected arrivals, learned from job history, to each hourly plan. CARMA is paired with a 1–24-hour carbon-intensity forecaster that combines direct gradient-boosted models with normalised conformal calibration and adaptive conformal inference. The methods are evaluated on three years of regional grid data for Great Britain with four sites and 52 out-of-sample weeks in 2024. The forecaster reduces mean absolute error by about 23% relative to persistence at horizons of 1–6 hours and keeps its upper bounds calibrated despite decarbonisation drift. With migration allowed, CARMA reduces emissions by 36–68% relative to carbon-agnostic operation, comes within 5.8–8.7% of the clairvoyant optimum (against 17.6–21.5% for myopic receding-horizon control), and lowers late work under heavy load. Anticipating demand is worth 10–13 percentage points of optimality gap, while perfect carbon foresight is worth 1–3 points to a myopic planner. Risk-averse conformal carbon

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pricing helps myopic planners but not CARMA. Without migration, reductions are smaller (9–18% at the optimum) and forecast accuracy becomes the binding constraint.

Keywords: Carbon-aware computing, Data centre scheduling, Model predictive control, Minimum-cost flow, Conformal prediction, Carbon intensity forecasting

1. Introduction

Data centres are among the fastest-growing electricity consumers. Efficiency gains kept their energy use nearly flat through the 2010s even as computing output grew several-fold [1, 2]. That buffer is now thinning, and information and communication technology accounts for an estimated 2.1–3.9% of global greenhouse-gas emissions [3]. Large machine-learning workloads have drawn particular attention [4, 5, 6]. The carbon intensity of grid electricity varies by a factor of ten or more across regions and hours, driven by the output of wind, solar and dispatchable plant [7]. Operators can therefore cut emissions without new hardware by moving flexible work to cleaner places and times. Such *carbon-aware computing* [8, 9] is an operations problem: capacity is finite, deadlines bind, and both the grid and the job stream are uncertain.

Batch jobs such as model training, analytics, rendering and backups are a natural target, because they tolerate some delay. Previous work has shown the potential of temporal shifting (running later) [10, 11, 12] and of spatial shifting (running elsewhere) [13, 14]. It has also shown their limits. Sukprasert et al. [15] found that practical reductions from spatio-temporal shifting fall well short of idealised bounds. Most methods are evaluated with a scheduler that reacts to the jobs it has already seen, using point forecasts of carbon intensity. Two questions have received less attention. First, how much is lost because an online scheduler does not know which jobs will arrive next? Second, does it pay to price the uncertainty of carbon-intensity forecasts into scheduling decisions?

This paper answers both questions with a new algorithm and a forecasting model, evaluated on three years of real regional grid data for Great Britain. It makes four contributions.

1. **Formulation.** We state multi-site, deadline-constrained, width-limited scheduling of deferrable bag-of-tasks jobs, including the emissions of

data transfer, as a minimum-cost flow problem (Proposition 1). Offline clairvoyant optima and every online planning step can therefore be computed exactly and quickly by network simplex. We also prove that myopic receding-horizon scheduling can be arbitrarily worse than the clairvoyant optimum even with perfect carbon forecasts (Proposition 2).

2. **Algorithm.** We propose CARMA (Carbon-aware Anticipatory Receding-horizon Multi-site Allocation). At every hour, CARMA adds *ghost jobs* to its planning network. These represent expected arrivals, learned from the operator’s history, so the planner reserves scarce low-carbon capacity for work that has not yet arrived. The planning problem stays a single min-cost flow and decisions take tens of milliseconds.
3. **Forecasting model.** We develop a probabilistic 1–24-hour carbon-intensity forecaster. It consists of direct gradient-boosted models with horizon-wise normalised conformal calibration and adaptive conformal inference, which holds coverage under the strong decarbonisation drift seen in 2024. Its calibrated quantiles let us test risk-adjusted carbon pricing inside the scheduler.
4. **Evidence.** Across 52 out-of-sample weeks, three load levels and two operating regimes, we compare CARMA with five implementable online baselines, clairvoyant and ablated variants, and the clairvoyant oracle. The results separate the value of anticipating demand from the value of better carbon forecasts, and show when each matters.

The results carry an operational message. Once spatial shifting is allowed, the dominant loss comes from demand myopia, not from forecast error. Anticipating demand cuts the gap to the clairvoyant optimum by a factor of 2.2–3.0. By contrast, perfect carbon foresight or risk-averse carbon pricing adds little once demand is anticipated. Without migration, temporal shifting alone achieves much smaller reductions. The forecast-driven planners are then within 4–6% of the optimum, and the remaining gap is due almost entirely to forecast error. The rest of the paper is organised as follows. Section 2 reviews related work. Section 3 presents the model, the forecaster, CARMA and the experimental design. Section 4 reports results, Section 5 discusses implications and limitations, and Section 6 concludes.

64 2. Related work

65 *Carbon-aware workload shifting.* Early green data-centre schedulers matched
66 deferrable jobs to on-site solar production, for example GreenSlot [16] and
67 GreenHadoop [11]. Other work exploited electricity-price and carbon differ-
68 ences across sites through geographical load balancing [17, 13] and dynamic
69 right-sizing [18]. With public carbon-intensity feeds, attention moved to grid
70 emissions. Wiesner et al. [10] quantified the reductions from temporal shift-
71 ing in several regions, including Great Britain, and proposed forecast-based
72 lowest-window scheduling. Google’s carbon-intelligent compute system sets
73 next-day virtual capacity curves from carbon and demand forecasts [8]. Car-
74 bonScaler varies the allocation of elastic jobs with carbon intensity [12]. Eco-
75 visor exposes the energy system to applications [19]. Carbon Explorer weighs
76 scheduling against investment in renewables and storage [20]. Load migra-
77 tion between data centres can absorb curtailed renewable generation [14],
78 and data-centre demand response has been surveyed in [21]. Sukprasert et
79 al. [15] bound the potential of spatio-temporal shifting across 123 regions and
80 find that simple policies capture most of the achievable benefit. Our study
81 complements theirs by identifying *which* information limits online schedulers
82 when capacity at clean sites is scarce.

83 *Energy-aware scheduling in operations.* Energy-efficient scheduling is a ma-
84 ture topic in manufacturing operations [22], and green scheduling models
85 have appeared in this journal, for example for parallel machines with job
86 splitting [23] and for cloud resource allocation [24]. Most such models are de-
87 terministic and solved offline, often with metaheuristics. We instead exploit
88 exact network-flow structure [25, 26] and embed it in an online lookahead
89 policy. In the taxonomy of Powell [27], CARMA is a deterministic lookahead
90 policy with a modified (anticipatory) model, in the spirit of model predictive
91 control [28, 29].

92 *Carbon-intensity forecasting.* Forecasts of grid carbon intensity use statistical
93 decomposition [30], machine learning with many exogenous drivers [31], day-
94 ahead regression for building demand response [32], and hierarchical models
95 of the generation mix, as in CarbonCast [33]. For Great Britain, NESO
96 publishes regional forecasts through its Carbon Intensity API [34]. These
97 studies focus on point accuracy. For decisions, calibrated uncertainty matters
98 as well. Conformal prediction [35, 36, 37] provides distribution-free intervals
99 and has been extended to quantile regression [38], to multi-horizon [39] and

dynamic time series [40], and to arbitrary distribution shift through adaptive conformal inference [41]. We use these tools to build a calibrated forecaster and to test, in a controlled way, whether risk-adjusted carbon costs improve scheduling. The question is motivated by the optimizer’s curse [42] and by the robust-optimisation view that point-estimate solutions can be fragile [43, 44].

3. Material and methods

3.1. Problem setting

We consider an operator that runs a set $\mathcal{S}$ of data-centre sites connected by a wide-area network. Time is divided into hourly slots $t = 0, 1, 2, \dots$. Deferrable batch jobs arrive online. Job j is released at the start of hour a_j at origin site $o_j \in \mathcal{S}$. It carries w_j server-hours of work made of independent tasks (a bag of tasks), can use at most m_j servers concurrently, and must finish before the start of hour $d_j > a_j$. Work may be paused and resumed, and any part of it may run at a site other than its origin. Remote execution requires moving input data. At site s and hour t , $K_s(t)$ servers are free for batch work once the interactive load has been served.

Let $I_s(t)$ be the average carbon intensity of the electricity supplied to site s during hour t (gCO₂/kWh). Executing one server-hour of job j at site s during hour t emits

$$c_{o_j s}(t) = e \text{PUE}_s I_s(t) + \mathbf{1}[s \neq o_j] \delta \eta \frac{1}{2} (I_{o_j}(t) + I_s(t)), \quad (1)$$

where e is the IT energy per server-hour, PUE_s is the power usage effectiveness of site s , δ is the data volume moved per migrated server-hour and η is the energy intensity of network transfer (kWh/GB). The transfer term charges network energy at the mean intensity of the two endpoints. With decision variables $x_{js}(t) \geq 0$ (server-hours of job j run at site s in hour t), the offline version of the problem is the linear programme

$$\min_{x, u \geq 0} \quad \sum_j \sum_s \sum_{t=a_j}^{d_j-1} c_{o_j s}(t) x_{js}(t) + \sum_j \Pi u_j \quad (2)$$

$$\text{s.t.} \quad \sum_s \sum_{t=a_j}^{d_j-1} x_{js}(t) + u_j = w_j \quad \forall j, \quad (3)$$

$$\sum_s x_{js}(t) \leq m_j \quad \forall j, \quad t \in [a_j, d_j), \quad (4)$$

$$\sum_{j: a_j \leq t < d_j} x_{js}(t) \leq K_s(t) \quad \forall s, t, \quad (5)$$

where u_j is work left unfinished at the deadline and Π is a lateness penalty much larger than any emission cost. When migration is not allowed, $x_{js}(t) = 0$ for $s \neq o_j$. With complete knowledge of arrivals and intensities, the optimum of (2)–(5) is a lower bound for every online policy, and we use it as the *clairvoyant oracle*. An online scheduler has to fix $x_{.s}(t)$ at the start of hour t . At that point it knows the jobs released so far, the current intensities $I_s(t)$ and nothing more than forecasts of $I_s(t+h)$ and of future arrivals.

Proposition 1 (Network-flow structure). *Problem (2)–(5) is a minimum-cost flow problem on a network with a source, one node per job, one node per (job, hour) pair, one node per (hour, site) pair and a sink. Arc capacities are w_j (source $\rightarrow j$), m_j ($j \rightarrow (j, t)$) and $K_s(t)$ ($(t, s) \rightarrow \text{sink}$). The only arcs with a cost are $(j, t) \rightarrow (t, s)$, priced at $c_{o_js}(t)$, and a slack arc $j \rightarrow \text{sink}$ priced at Π .*

Proof. Conservation at the job node is Eq. (3). The capacity of arc $j \rightarrow (j, t)$ is Eq. (4), conservation at (j, t) splits that hour’s work across sites, and the capacity of $(t, s) \rightarrow \text{sink}$ is Eq. (5). Conversely, every feasible flow defines a feasible (x, u) with the same cost. The constraint matrix of the flow formulation is a node-arc incidence matrix, in which every column has exactly one +1 and one −1, so it is totally unimodular. $\square$

Proposition 1 has a practical consequence. The oracle and every planning problem solved online (Section 3.3) can be solved exactly by network simplex [25], and integral data (work quantised to 0.01 server-hours) give integral optimal flows. We use the OR-Tools implementation [45]. On the instances studied, network simplex returned the same objective value as a general-purpose LP solver (HiGHS) to within rounding, and it cut the total run time of the online planners by a factor of about six.

3.2. Probabilistic carbon-intensity forecasting

Point model. For each horizon $h = 1, \dots, H$ ($H = 24$ h) we fit a direct gradient-boosted regression tree model [46] that predicts the change $I_s(t+h) - I_s(t)$ from information available at issue time t . Direct per-horizon models avoid the error accumulation of recursive strategies [47]. One model per horizon is pooled across sites, with the site as a categorical feature. The features are the current intensity; differences between the current value and lags of 1, 2, 3, 6, 12 and 23 hours; the value observed at the same hour of the target on the previous day and in the previous week; a 24-hour rolling

mean and mean absolute change (volatility); the national (GB) intensity; the regional wind, solar and gas shares; and calendar encodings of the target time (hour of day, day of week, annual sine and cosine). Models are fitted with absolute-error loss, so they estimate the conditional median. We use scikit-learn’s histogram gradient boosting [48] with 400 iterations, learning rate 0.05, 31 leaves and at least 50 samples per leaf. The models use only information available when the forecast is issued, and no weather forecasts. This is deliberate: the goal is a forecaster any operator can build from a public intensity feed.

Conformal upper quantiles. A point forecast $\hat{I}_s(t+h|t)$ is turned into calibrated upper quantiles by split conformal prediction [35, 36] with normalised scores. On a calibration window disjoint from training, we compute for each site s and horizon h

$$r_{s,h}(t) = \frac{I_s(t+h) - \hat{I}_s(t+h|t)}{\sigma_s(t) + \epsilon}, \quad (6)$$

where $\sigma_s(t)$ is the 24-hour mean absolute hourly change observed at issue time and $\epsilon = 5$ gCO₂/kWh. For a target level β , the upper forecast is $U_s^\beta(t+h|t) = \hat{I}_s(t+h|t) + \hat{q}_{s,h}(\beta_{s,h}(t))(\sigma_s(t) + \epsilon)$, where $\hat{q}_{s,h}(\cdot)$ is the empirical quantile function of the calibration scores. Grid carbon intensity is not exchangeable over time: 2024 was markedly cleaner than 2022–2023 (Fig. 2). We therefore update the working level online by adaptive conformal inference (ACI) [41]. Once $I_s(t+h)$ is observed,

$$\alpha_{s,h} \leftarrow \alpha_{s,h} + \gamma((1-\beta) - \mathbf{1}[I_s(t+h) > U_s^\beta(t+h|t)]), \quad \beta_{s,h}(t) = 1 - \alpha_{s,h}, \quad (7)$$

with step size $\gamma = 0.005$. By the ACI result of [41], the long-run frequency with which realised intensities exceed U^β converges to $1 - \beta$ for any data sequence, with no distributional assumption. This matters for the scheduler: the level β becomes an interpretable risk dial, and it stays valid under drift. With $\beta = 0.5$ the procedure also removes level shifts in the point model (a recalibrated median).

3.3. CARMA: carbon-aware anticipatory receding-horizon multi-site allocation

A standard receding-horizon (model predictive control, MPC) scheduler [28] solves, at every hour t , the flow problem of Proposition 1 restricted to pending

191 jobs and the window $[t, t + H)$. Future intensities are replaced by forecasts,
 192 and only the first hour of the plan is executed. Such a planner is *myopic*
 193 *about demand*: it sees only jobs that have already arrived. The next result
 194 shows that this can be arbitrarily costly even when carbon intensities are
 195 known exactly.

196 **Proposition 2** (Myopia can be unboundedly costly). *Even with perfect in-*
 197 *tensity forecasts, myopic receding-horizon scheduling has no finite competitive*
 198 *ratio with respect to the clairvoyant oracle.*

199 *Proof.* Take a clean site A with one server and a dirty site B with ample
 200 capacity. Let $c_A(0) = \varepsilon$, $c_A(1) = 0$ and $c_B(0) = c_B(1) = c$, and ignore
 201 transfer costs. A flexible job ($w = 1$, $m = 1$) arrives at $t = 0$ with deadline
 202 $d = 2$. An urgent job ($w = 1$) arrives at $t = 1$ with $d = 2$. At $t = 0$ the
 203 myopic planner sees only the flexible job and defers it to A at $t = 1$ (cost
 204 $0 < \varepsilon$). At $t = 1$ both jobs must run in the same hour, one of them at B , so
 205 the total cost is c . The oracle runs the flexible job at $t = 0$ and the urgent
 206 job at $t = 1$, both at A , for a total of ε . The ratio c/ε is unbounded as
 207 $\varepsilon \rightarrow 0$. $\square$

208 The mechanism in Proposition 2 is common in practice. Low-carbon
 209 capacity is scarce, and a myopic planner spends it on flexible work that
 210 could have waited or run elsewhere, leaving urgent jobs that arrive later
 211 to run on carbon-intensive capacity. CARMA addresses it by anticipating
 212 demand inside the same flow problem, and optionally prices forecast risk.

213 *Anticipated demand (ghost jobs).* From the operator's own history we esti-
 214 mate, for every hour of the week τ , origin o and window class b , the expected
 215 work $\bar{W}_{\tau ob}$ and the expected total width $\bar{M}_{\tau ob}$ of arriving jobs. Window classes
 216 are the bins $\mathcal{B} = \{1, 2, 4, 6, 9, 13, 18, 24\}$ hours. A job is assigned to the largest
 217 bin not exceeding its window, so ghost jobs are never more flexible than
 218 the jobs they represent. At hour t , for each $k = 1, \dots, \ell$ (the look-ahead),
 219 CARMA adds to the planning problem a *ghost job* for every (o, b) with work
 220 $\kappa \bar{W}_{\tau(t+k)ob}$, width $\kappa \bar{M}_{\tau(t+k)ob}$, release $t + k$ and deadline $\min(t + k + b, t + H)$.
 221 Ghost jobs compete with real jobs for capacity, but their slack arc is priced
 222 at a moderate penalty Π_g , set to 1.5 times the largest local unit cost in the
 223 window, rather than at Π . They can therefore reserve clean capacity when it
 224 is economical but never force real work to be late. The weight $\kappa \geq 0$ scales
 225 how much of the expected demand is anticipated. With $\kappa = 0$, CARMA
 226 reduces to myopic MPC.

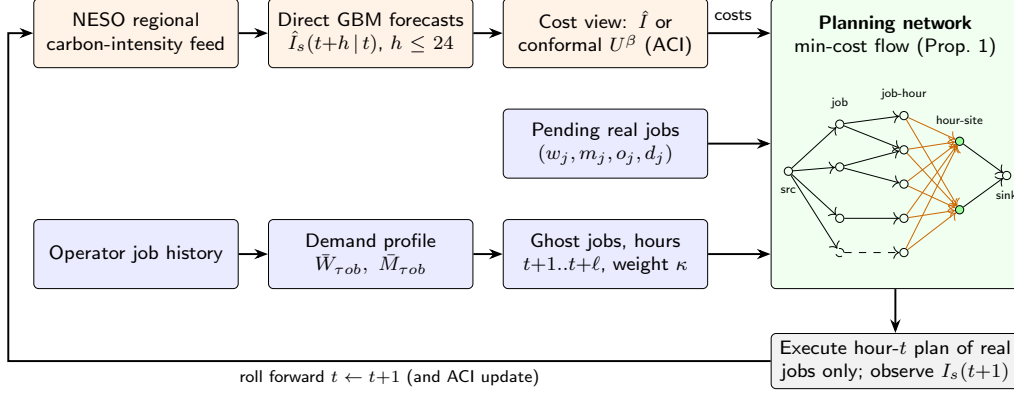


Figure 1: The CARMA decision loop. Every hour, carbon-intensity forecasts price the (job-hour)→(hour-site) arcs, and pending real jobs and ghost jobs for expected arrivals (dashed) share the planning network. The min-cost flow is solved exactly, and only the first-hour allocation of real jobs is executed.

227 *Carbon cost view.* Future-hour costs in Eq. (1) are evaluated at a forecast of
 228 $I_s(t+h)$; for the current hour ($h = 0$) the observed intensity is used. CARMA
 229 admits two cost views: the point forecast $\hat{I}$, or the conformal upper quantile
 230 U^β of Section 3.2. The second is motivated by the optimizer’s curse [42]:
 231 selecting the minimum over noisy estimates is biased towards optimism, and a
 232 level $\beta > 0.5$ penalises uncertain future hours in proportion to their calibrated
 233 spread. Whether this risk premium helps is an empirical question, and we
 234 treat the cost view as a hyperparameter.

235 *Algorithm.* Fig. 1 summarises the loop. At each hour t , CARMA (i) observes
 236 $I_s(t)$, the free capacity and the pending jobs; (ii) builds the cost cube $c_{os}(t+h)$
 237 for $h < H$ from the chosen forecast view; (iii) adds the ghost jobs for
 238 hours $t+1, \dots, t+\ell$; (iv) solves the resulting min-cost flow problem exactly;
 239 (v) executes the first-hour allocations of the real jobs only; and (vi) after
 240 observing the realised intensities, updates the ACI levels by Eq. (7) when
 241 conformal costs are used. Overdue work gets a one-hour window and is
 242 served with priority. The planning network has $O((n_t + \ell|\mathcal{S}||\mathcal{B}|)H|\mathcal{S}|)$ arcs,
 243 where n_t is the number of pending jobs, so each decision takes a few tens of
 244 milliseconds (Section 4). The hyperparameters (κ, ℓ) and the cost view are
 245 tuned on validation weeks that precede the test period (Section 4.2).

246 3.4. Baselines

247 We compare CARMA with the following online policies and the oracle.
 248 **ASAP-Local** is carbon-agnostic: it runs work at its origin as soon as pos-
 249 sible, earliest deadline first (EDF). **Greedy-Spatial** also runs work imme-
 250 diately, but at the site that is currently cleanest; this is geographical load
 251 balancing without temporal shifting [13]. **Forecast-Reserve** books, when
 252 a job arrives, the cheapest forecast (site, hour) slots in its window that still
 253 have capacity, and never revises the booking. This is the lowest-window
 254 strategy of [10] extended to multiple sites. **MPC** is the myopic receding-
 255 horizon flow scheduler with point forecasts. **MPC+conformal** uses the
 256 conformal upper quantile at the level that was best for MPC on validation
 257 data. **MPC-Perfect** is MPC given the realised intensities, which isolates
 258 the value of forecast accuracy. **CARMA-Perfect** is CARMA given the
 259 realised intensities, and **CARMA+conformal** is CARMA with the best
 260 conformal cost view. **Oracle** solves (2)–(5) with full knowledge of arrivals
 261 and intensities.

262 3.5. Data and experimental design

263 *Carbon intensity.* We use half-hourly regional carbon intensity for Great
 264 Britain from the Carbon Intensity API of the National Energy System Op-
 265 erator (NESO) [34] from 1 January 2022 to 31 December 2024. Values are
 266 averaged to hourly resolution. Four regions host the candidate sites: South
 267 Scotland, North West England, West Midlands and London. They span the
 268 north–south gradient of the British grid, from wind- and nuclear-rich South
 269 Scotland to gas-dependent London, and include the main data-centre mar-
 270 kets of London, Manchester (North West England) and Birmingham (West
 271 Midlands). Gaps (602 of 26,304 hours in at least one region) were filled by
 272 linear interpolation. The period up to 30 June 2023 is used to train the
 273 forecaster, July–September 2023 to calibrate the conformal scores, October–
 274 December 2023 to tune the scheduler, and all 52 weeks starting on Mondays
 275 in 2024 as the out-of-sample test set.

276 *Sites and energy parameters.* Batch capacities are 300, 400, 400 and 600
 277 servers for South Scotland, North West England, West Midlands and Lon-
 278 don. This reflects a small, clean northern site and a large site in London.
 279 Interactive load takes $20\% \pm 15\%$ of each site’s capacity on a diurnal cy-
 280 cle that peaks in the afternoon. We assume $e = 0.4$ kWh per server-hour.
 281 The assumed PUE values are 1.15, 1.20, 1.25 and 1.30 from north to south,

reflecting free-cooling potential. Migrated work moves $\delta = 2$ GB per server-hour at $\eta = 0.02$ kWh/GB. This value lies within the wide range of published estimates of transmission energy intensity [49]. We vary η in a sensitivity analysis.

Workloads. Job arrivals follow a non-homogeneous Poisson process with a diurnal peak in the afternoon and 30% lower weekend rates. Jobs originate at the four sites with probabilities 0.10, 0.20, 0.25 and 0.45. Width m_j is log-uniform on $[4, 64]$ servers and the run time at full width r_j is uniform on $\{1, \dots, 6\}$ hours, so $w_j = m_j r_j$. A share v of jobs is urgent (deadline $= a_j + r_j$, no temporal slack); for the others the window is $\lceil f_j r_j \rceil$ hours with $f_j \sim U[1.5, 4]$, capped at 24 hours. The baseline value is $v = 0.3$. The arrival rate is scaled so that the offered load equals a fraction $\rho \in \{0.3, 0.5, 0.7\}$ of the mean free capacity. Each test episode covers one week of arrivals. We also simulate 24 hours of arrivals before and after the week, so the evaluated week starts and ends under steady-state congestion. Metrics are computed only for jobs released during the week. For each week and load level we draw one workload trace with a fixed seed, and every policy runs on that same trace. The demand profile used by CARMA is estimated from 13 independent historical traces. We study two regimes: *spatio-temporal*, where migration is allowed, and *temporal-only*, where it is not.

Metrics and statistics. The primary metric is operational emissions (tCO₂) per week. We report the reduction relative to ASAP-Local and the optimality gap relative to the oracle, the share of work finished late, the share of work executed away from its origin, and the decision time. Policies are compared by paired two-sided Wilcoxon signed-rank tests over the 52 test weeks, with Holm correction across comparisons.

4. Results

4.1. Carbon-intensity data and forecasts

Fig. 2 shows the regional intensities. Over 2022–2024, mean intensity was 37 gCO₂/kWh in South Scotland, 58 in North West England, 156 in the West Midlands and 163 in London. At any hour the gap between the cleanest and dirtiest site is therefore large, which is what makes spatial shifting attractive. Intensities fell steadily: London’s annual mean dropped from 202 gCO₂/kWh in 2022 to 124 in 2024. Diurnal profiles peak in the early evening (around

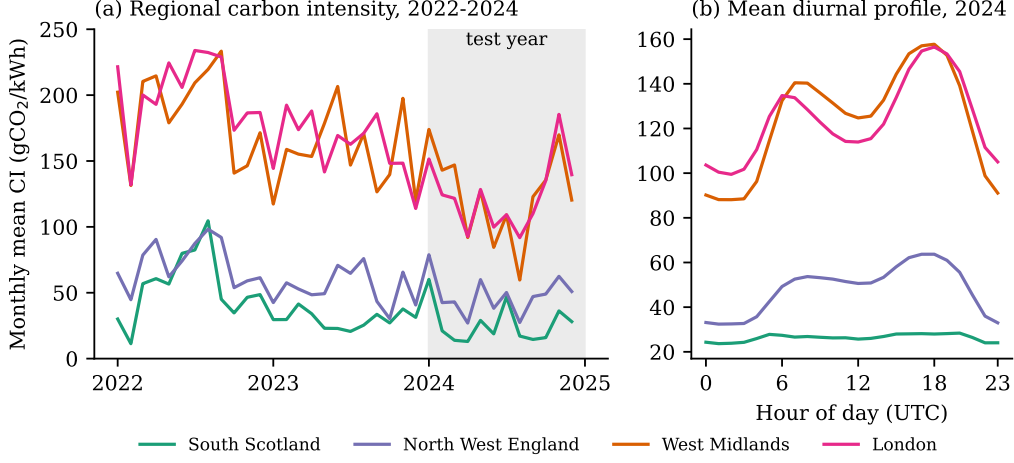


Figure 2: Hourly carbon intensity of the four candidate sites (NESO regional data). (a) Monthly means, 2022–2024; the shaded area is the out-of-sample test year. (b) Mean diurnal profile in 2024 (UTC).

18:00 UTC), and the profile of South Scotland is much flatter than those of the southern sites. The four series are positively correlated (pairwise correlations between 0.50 and 0.87), because national wind output drives all of them.

Table 1 and Fig. 3a summarise forecast accuracy in 2024. The forecaster was trained on 1.24 million (issue time, site, horizon) samples in about one minute. Averaged over horizons and sites, the direct GBM forecaster has a mean absolute error (MAE) of 31.2 gCO₂/kWh, against 34.4 for persistence and 40.4 for the seasonal-naïve forecast. The advantage is largest at short and medium horizons: MAE falls by 23% at 1 hour (6.2 against 8.0) and by 23% at 6 hours (23.3 against 30.2). Beyond about 18 hours, all methods converge to the seasonal-naïve level, and at 24 hours the GBM is slightly worse than the seasonal-naïve benchmark. Without weather forecasts, day-ahead changes in wind output are hard to predict from the intensity history alone [33]. By site, the GBM is most accurate for the volatile southern sites. In South Scotland, where intensity is low and often flat, persistence is marginally better.

Calibration matters more than point accuracy for the risk-adjusted cost view, and Fig. 3b shows the effect of distribution shift. Static split-conformal bounds, calibrated on July–September 2023, under-cover in 2024 at every

Table 1: Forecast accuracy on the 2024 test year (MAE in gCO₂/kWh; mean over sites unless stated).

Model	Horizon h (hours)					All horizons	
	1	3	6	12	24	MAE	RMSE
Persistence	8.0	19.3	30.2	37.4	40.4	34.4	48.4
Seasonal naive (24 h)	40.4	40.4	40.4	40.4	40.4	40.4	57.0
Direct GBM (this study)	6.2	14.9	23.3	33.9	42.4	31.2	41.0

MAE by site over all horizons (persistence / GBM): South Scotland 14.1 / 16.2; North West England 24.7 / 21.8; West Midlands 56.1 / 49.3; London 42.7 / 37.6.

level: at $\beta = 0.9$, annual coverage was 0.858 and monthly coverage ranged from 0.68 to 0.95. With ACI, annual coverage matches the target to within 0.002 at all levels (0.500, 0.599, 0.698, 0.798 and 0.898). Monthly coverage stays within ± 4 percentage points of the target, and coverage is uniform across horizons. At $\beta = 0.5$, the ACI-adjusted median lies on average 9.9 gCO₂/kWh below the point forecast, which reflects the downward drift in intensity that the static model does not capture.

4.2. Hyperparameter tuning

Fig. 4 reports the validation results (13 weeks, October–December 2023, $\rho \in \{0.3, 0.5, 0.7\}$, migration allowed). Anticipating demand helps across the whole grid: every tested configuration beats myopic MPC, whose mean gap to the oracle is 20.3%. The gap is smallest around $\kappa = 1$, where expected demand is anticipated without inflation. For $\ell = 12$ it falls to 7.3%, and for $\ell = 18$ to 7.2%. Over-anticipation ($\kappa = 1.25$ with $\ell = 18$) reserves too much clean capacity and is penalised. We selected $\kappa = 1$ and $\ell = 12$ hours. Configurations within 0.25 percentage points of the best were resolved towards the shorter look-ahead, which keeps the network smaller.

The cost view shows a clear interaction with anticipation (Fig. 4b). For myopic MPC, pessimistic conformal costs help substantially. The gap falls monotonically from 23.1% at $\beta = 0.5$ to 16.2% at $\beta = 0.8$, well below the 20.3% obtained with point forecasts. For CARMA, every conformal level is worse than the point forecast: the best level, $\beta = 0.6$, gives 8.2% against 7.3%. The explanation is that a risk premium on future hours makes the planner run work now rather than later, which keeps future clean capacity free. For a myopic planner this is a crude form of anticipation. Once CARMA

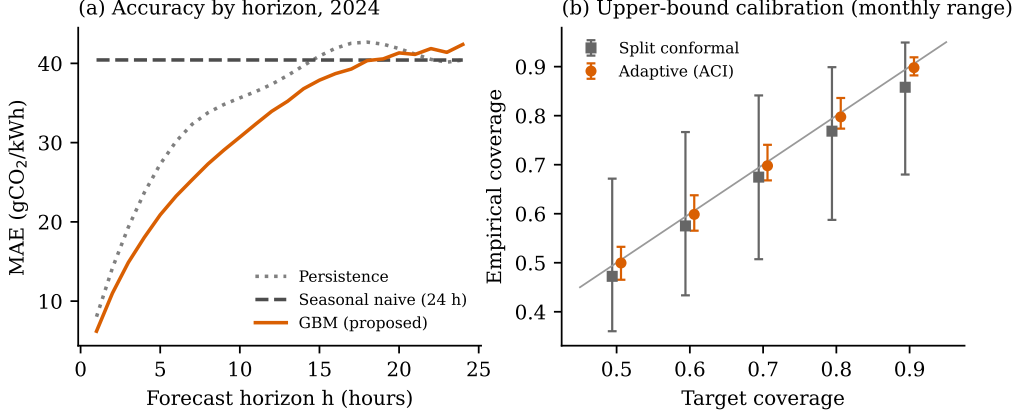


Figure 3: Forecast evaluation in 2024. (a) MAE by horizon, averaged over sites. (b) Empirical coverage of one-sided upper bounds U^β against the target level β for static split conformal calibration and adaptive conformal inference (ACI). Markers show the annual coverage and bars the range across the 12 months.

361 represents future demand explicitly, the same premium double-counts it and
362 gives up useful deferrals. The selected CARMA therefore uses point forecasts.
363 For the test we keep MPC+conformal at $\beta = 0.8$ and CARMA+conformal
364 at $\beta = 0.6$ as ablations.

365 4.3. Out-of-sample scheduling performance

366 Tables 2 and 3 report results for the 52 test weeks of 2024. Weekly traces
367 contain on average 920, 1,529 and 2,144 jobs (69,900, 115,600 and 162,600
368 server-hours) at $\rho = 0.3, 0.5$ and 0.7 . Fig. 5 shows the gaps to the oracle
369 with 95% confidence intervals.

370 *Spatio-temporal regime.* Carbon-aware scheduling gives large reductions when
371 migration is allowed. The oracle cuts emissions relative to ASAP-Local by
372 70.5%, 56.7% and 39.5% at $\rho = 0.3, 0.5$ and 0.7 . The achievable reduction
373 shrinks with load because less clean capacity is left per unit of work.
374 Greedy-Spatial realises only part of this potential (gaps of 35–65%), because
375 it has no temporal flexibility and floods the cleanest site with whatever ar-
376 rives first. Forecast-Reserve and MPC come closer, with gaps of 18–26%.
377 CARMA reduces emissions by 68.1%, 53.3% and 36.1% and reaches gaps of
378 8.7%, 8.2% and 5.8%. Relative to MPC this is a further emission reduction
379 of 7.6%, 9.4% and 8.8%, and the gap to the optimum shrinks by a factor

Table 2: Out-of-sample results in the spatio-temporal regime (migration allowed), 52 weeks of 2024. Emis.: mean weekly operational emissions (tCO₂). Red.: reduction relative to ASAP-Local (%). Gap: excess over the clairvoyant oracle (%). Late: share of work finished after its deadline (%). Mig.: share of work executed away from its origin (%). p : Holm-adjusted two-sided Wilcoxon signed-rank test of weekly emissions against CARMA. The best gap among implementable online policies is in bold.

Load	Policy	Emis.	Red.	Gap	Late	Mig.	p
$\rho = 0.3$	ASAP-Local	3.73	0.0	274.7	0.00	0.0	<0.001
	Greedy-Spatial	1.77	53.8	64.6	0.00	81.6	<0.001
	Forecast-Reserve	1.40	63.7	25.7	0.00	83.7	<0.001
	MPC	1.34	65.5	18.9	0.00	69.2	<0.001
	MPC+conformal	1.29	66.7	14.2	0.00	70.0	<0.001
	MPC-Perfect	1.30	66.4	15.9	0.00	68.9	<0.001
	CARMA	1.23	68.1	8.7	0.00	69.9	–
	CARMA+conformal	1.24	68.1	8.9	0.00	70.2	0.122
	CARMA-Perfect	1.17	70.0	1.8	0.00	69.6	<0.001
	Oracle	1.15	70.5	0.0	0.00	70.0	–
$\rho = 0.5$	ASAP-Local	6.12	0.0	142.7	0.05	0.0	<0.001
	Greedy-Spatial	4.15	33.3	58.8	0.00	76.3	<0.001
	Forecast-Reserve	3.36	46.6	25.0	0.02	78.2	<0.001
	MPC	3.27	48.1	21.4	0.01	50.0	<0.001
	MPC+conformal	3.18	49.7	17.4	0.00	50.7	<0.001
	MPC-Perfect	3.22	48.9	19.5	0.02	49.4	<0.001
	CARMA	2.96	53.3	8.2	0.00	50.3	–
	CARMA+conformal	2.98	53.2	8.6	0.00	50.4	0.001
	CARMA-Perfect	2.83	55.7	2.3	0.00	50.0	<0.001
	Oracle	2.77	56.7	0.0	0.00	50.7	–
$\rho = 0.7$	ASAP-Local	8.32	0.0	68.1	6.50	0.0	<0.001
	Greedy-Spatial	6.79	19.1	35.3	0.00	73.4	<0.001
	Forecast-Reserve	6.15	26.9	21.7	1.16	73.6	<0.001
	MPC	5.95	29.3	17.6	2.00	31.6	<0.001
	MPC+conformal	5.85	30.7	15.3	0.78	32.1	<0.001
	MPC-Perfect	5.89	30.1	16.3	2.08	31.4	<0.001
	CARMA	5.43	36.1	5.8	0.05	31.5	–
	CARMA+conformal	5.45	35.9	6.2	0.02	31.6	<0.001
	CARMA-Perfect	5.26	38.4	1.9	0.05	31.1	<0.001
	Oracle	5.18	39.5	0.0	0.00	30.7	–

Table 3: Out-of-sample results in the temporal-only regime (no migration), 52 weeks of 2024. Columns as in Table 2. Greedy-Spatial is omitted because without migration it coincides with ASAP-Local.

Load	Policy	Emis.	Red.	Gap	Late	p
$\rho = 0.3$	ASAP-Local	3.73	0.0	22.0	0.00	<0.001
	Forecast-Reserve	3.30	11.9	7.1	0.36	<0.001
	MPC	3.26	13.1	5.7	0.01	0.367
	MPC+conformal	3.40	9.2	10.4	0.00	<0.001
	MPC-Perfect	3.11	17.5	0.2	0.02	<0.001
	CARMA	3.26	13.1	5.7	0.00	–
	CARMA+conformal	3.27	12.9	5.9	0.00	0.020
	CARMA-Perfect	3.11	17.6	0.1	0.01	<0.001
	Oracle	3.11	17.7	0.0	0.00	–
$\rho = 0.5$	ASAP-Local	6.12	0.0	17.2	0.05	<0.001
	Forecast-Reserve	5.59	8.9	6.6	3.15	<0.001
	MPC	5.52	10.2	5.1	0.64	0.219
	MPC+conformal	5.67	7.7	8.0	0.10	<0.001
	MPC-Perfect	5.35	13.4	1.3	0.88	<0.001
	CARMA	5.52	10.3	5.0	0.19	–
	CARMA+conformal	5.54	10.1	5.2	0.11	0.003
	CARMA-Perfect	5.31	14.1	0.4	0.26	<0.001
	Oracle	5.29	14.5	0.0	0.01	–
$\rho = 0.7$	ASAP-Local	8.32	0.0	9.5	6.50	<0.001
	Forecast-Reserve	7.96	4.2	4.8	8.35	<0.001
	MPC	7.91	4.9	4.1	12.99	0.244
	MPC+conformal	8.00	3.9	5.2	8.09	<0.001
	MPC-Perfect	7.81	6.3	2.6	15.31	<0.001
	CARMA	7.92	4.8	4.2	6.86	–
	CARMA+conformal	7.95	4.6	4.5	6.72	0.004
	CARMA-Perfect	7.78	6.7	2.1	7.85	<0.001
	Oracle	7.63	8.6	0.0	0.96	–

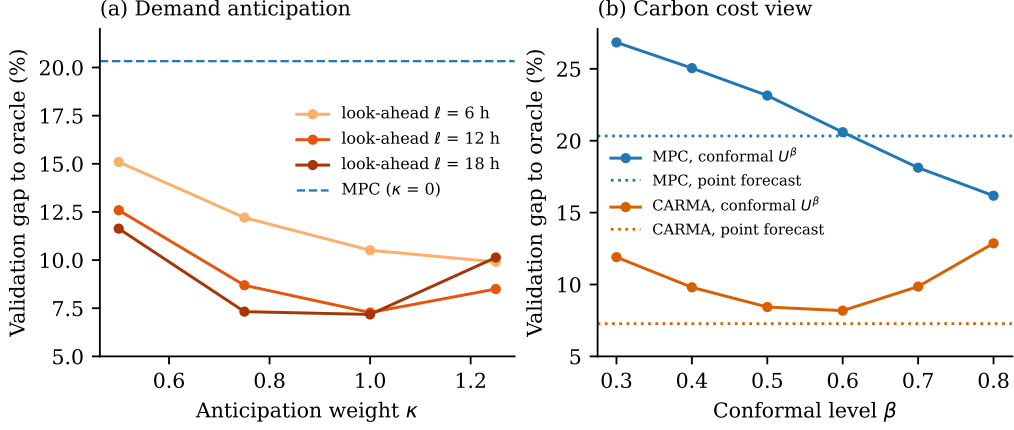


Figure 4: Validation results (2023-Q4, mean over 13 weeks and three load levels). (a) Gap to the oracle as a function of the anticipation weight κ and look-ahead ℓ , using point forecasts. (b) Effect of the carbon cost view: conformal upper quantiles U^β against point forecasts, for myopic MPC and for CARMA ($\kappa = 1$, $\ell = 12$).

of 2.2–3.0. CARMA’s emissions are lower than MPC’s in 51, 52 and 52 of the 52 weeks, and lower than Forecast-Reserve’s in every week (Fig. 6b). All differences between CARMA and the online baselines are significant after Holm correction ($p < 0.001$). The only exception is CARMA+conformal at $\rho = 0.3$ ($p = 0.12$). CARMA also finishes more work on time under heavy load: at $\rho = 0.7$, 0.05% of its work is late, against 2.0% for MPC, 1.2% for Forecast-Reserve and 6.5% for ASAP-Local. Reserving clean capacity for urgent arrivals also reserves capacity in general. CARMA migrates about the same share of work as MPC (31–70%), so its advantage comes from better timing of the use of scarce clean capacity, not from moving more data. For a fleet of 1,700 batch servers at $\rho = 0.5$, the difference between CARMA and MPC is 0.31 tCO₂ per week. The difference from carbon-agnostic operation is 3.16 tCO₂ per week.

Value of information. The clairvoyant variants separate two sources of loss. With perfect intensity forecasts, the myopic MPC-Perfect still has gaps of 15.9%, 19.5% and 16.3%. Perfect carbon foresight is thus worth only 1.3–3.0 percentage points to a myopic planner. Anticipating demand is worth 10.3–13.3 points (MPC versus CARMA). With both, CARMA-Perfect comes within 1.8–2.3% of the oracle. The two kinds of information are therefore complementary. Once demand is anticipated, forecast accuracy is worth 3.9–

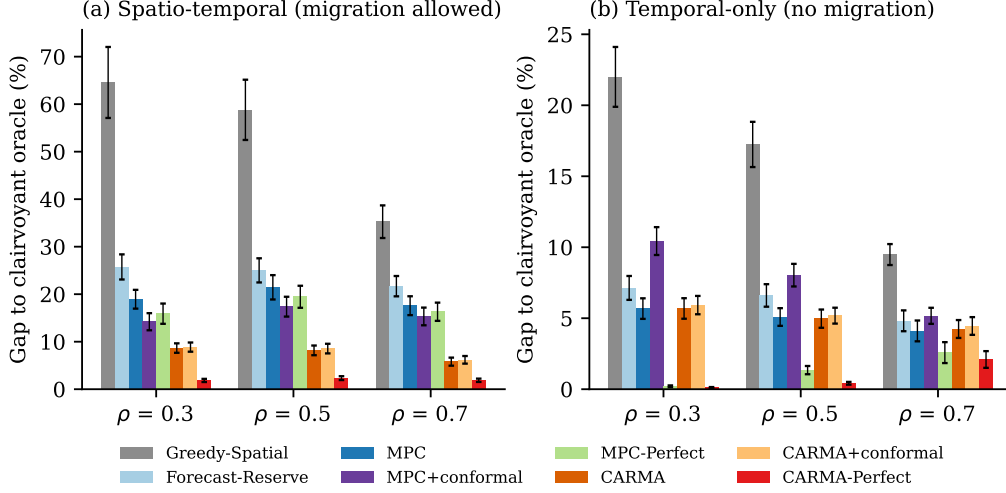


Figure 5: Mean gap to the clairvoyant oracle over the 52 test weeks of 2024 with 95% confidence intervals. ASAP-Local is omitted (gap 68–275% with migration, 9–22% without).

400 6.8 points (CARMA versus CARMA-Perfect), two to three times more than
 401 for the myopic planner. A myopic planner wastes clean capacity whatever
 402 its forecasts, so it cannot benefit from better ones. The weekly results in
 403 Fig. 6a show that the ranking is stable across seasons, although the level of
 404 achievable reductions varies from week to week with wind conditions.

405 *Temporal-only regime.* Without migration (Table 3), each job can only move
 406 in time at its origin, and the picture changes. Achievable reductions are
 407 much smaller: the oracle saves 17.7%, 14.5% and 8.6%. MPC and CARMA
 408 are statistically indistinguishable here ($p \geq 0.22$), both about 4–6% from the
 409 oracle, and both are better than Forecast-Reserve. Anticipation brings no
 410 benefit because there is no shared pool of clean capacity to protect. Each
 411 site’s capacity serves only its own jobs, and the temporal windows of flex-
 412 ible jobs rarely collide with urgent arrivals in a way that deferral cannot
 413 absorb. In this regime the remaining loss is due almost entirely to forecast
 414 error. MPC-Perfect and CARMA-Perfect close the gap to 0.1–2.6%, and
 415 risk-averse conformal costs make MPC worse (gaps of 5.2–10.4%), because
 416 they suppress deferrals that would have paid off. Under heavy load ($\rho = 0.7$),
 417 temporal shifting also raises lateness: the capacity at the origin sites is itself
 418 insufficient, and 6.5% of work is late even under ASAP-Local. CARMA’s

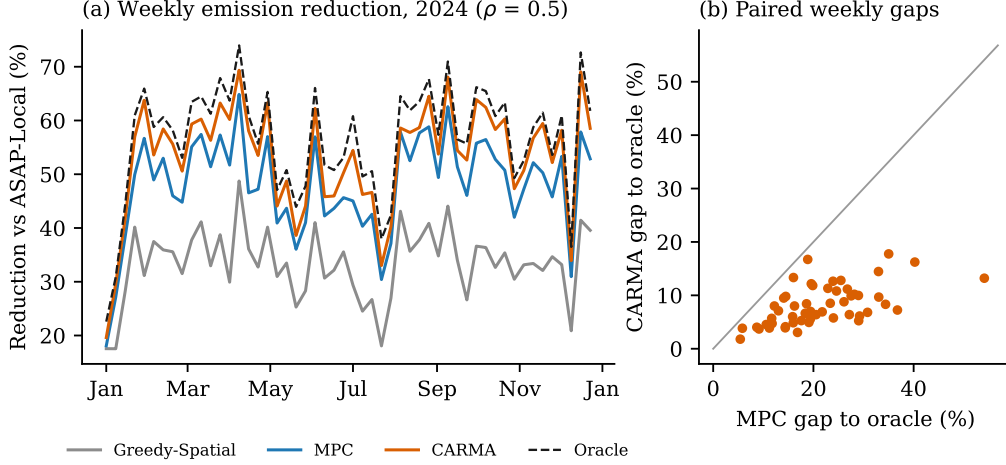


Figure 6: Weekly results in 2024 (spatio-temporal regime, $\rho = 0.5$). (a) Emission reduction relative to ASAP-Local. (b) Paired weekly gaps to the oracle; points below the diagonal are weeks in which CARMA outperforms MPC.

late share (6.9%) is close to that level, whereas MPC’s is 13.0%.

Computation. A CARMA decision takes 20–47 ms on one CPU core, against 2–7 ms for MPC. The difference comes from the ghost jobs, which enlarge the planning network. This is negligible for hourly decisions. A clairvoyant weekly oracle with up to about 2,700 jobs (including warm-up and trailing load) is solved exactly in under one second.

4.4. Sensitivity analysis

Table 4 varies one factor at a time around the base case: spatio-temporal regime, $\rho = 0.5$, $v = 0.3$, $\eta = 0.02$ kWh/GB. Each setting uses 26 alternating test weeks of 2024 with fresh workload seeds. CARMA keeps its hyperparameters from the base case. CARMA has the smallest gap of all online policies in every setting. Its weekly emissions are below MPC’s in 155 of the 156 setting-weeks, and every comparison with a baseline is significant (all $p < 10^{-7}$).

Share of urgent jobs. The benefit of anticipation grows with the share of urgent work, because urgent arrivals are precisely the demand that myopic planners crowd out. With $v = 0.5$, CARMA’s gap is 5.8% against 20.3% for MPC, and it has no late work (MPC 0.15%). With $v = 0.1$, the gap is 10.4%

Table 4: Sensitivity analysis (spatio-temporal regime; 26 alternating weeks of 2024 per setting; base case $\rho = 0.5$, $v = 0.3$, $\eta = 0.02$ kWh/GB). Oracle red.: emission reduction of the oracle relative to ASAP-Local (%). Gap columns: excess over the oracle (%). Wins: weeks in which CARMA emits less than MPC. Late: share of late work (%), MPC / CARMA.

Setting	Oracle red.	Gap to oracle (%)				Wins vs		Late (%)
		Greedy-Sp.	F.-Reserve	MPC	CARMA	MPC	MPC / CARMA	
Urgent share $v = 0.1$	60.3	74.8	26.3	20.9	10.4	26/26		0.00 / 0.00
Urgent share $v = 0.5$	55.0	51.1	24.5	20.3	5.8	26/26		0.15 / 0.00
Network energy $\eta = 0.005$ kWh/GB	60.7	66.2	26.6	25.0	9.4	26/26		0.01 / 0.00
Network energy $\eta = 0.06$ kWh/GB	52.3	62.4	27.6	18.1	7.4	26/26		0.00 / 0.00
Load $\rho = 0.4$, profile learned at 0.5	65.6	72.8	27.0	21.3	10.8	25/26		0.00 / 0.00
Load $\rho = 0.6$, profile learned at 0.5	49.3	52.2	25.7	22.1	6.9	26/26		0.34 / 0.01

437 against 20.9%. When almost all work is flexible, there is less urgent demand
438 to protect capacity for.

439 *Network energy intensity.* The ranking is unchanged over a twelve-fold range
440 of transmission energy intensity. Cheaper transfers ($\eta = 0.005$ kWh/GB)
441 raise the oracle’s reduction to 60.7%, and CARMA’s gap is 9.4% against
442 25.0% for MPC. Expensive transfers ($\eta = 0.06$ kWh/GB) lower the oracle’s
443 reduction to 52.3%, with gaps of 7.4% and 18.1%. The share of migrated
444 work changes only modestly (49–51% for CARMA), because the intensity
445 difference between South Scotland and London is far larger than any plausible
446 transfer overhead.

447 *Misspecified demand profile.* CARMA’s ghost jobs come from a profile learned
448 at $\rho = 0.5$. When the actual load is 20% higher ($\rho = 0.6$), so that CARMA
449 under-anticipates, its gap is 6.9% against 22.1% for MPC, and its late share
450 is 0.01% against 0.34%. When the actual load is 20% lower ($\rho = 0.4$), so that
451 it over-anticipates, CARMA reserves more clean capacity than needed. Its
452 gap rises to 10.8%, but that is still half of MPC’s 21.3%, and CARMA wins
453 in 25 of the 26 weeks. Anticipation is therefore robust to moderate errors
454 in the demand profile in both directions. The operational safeguard is the
455 softer penalty on ghost jobs: the planner never accepts lateness of real work
456 in exchange for serving ghost demand.

457 5. Discussion

458 5.1. Managerial implications

459 *Protect scarce clean capacity.* The largest operational loss in multi-site carbon-
460 aware scheduling does not come from imperfect carbon forecasts. It comes
461 from spending low-carbon capacity on work that could have waited or run
462 elsewhere. Proposition 2 shows this in the worst case, and the 2024 experi-
463 ments confirm it empirically: with migration allowed, a myopic planner with
464 perfect foresight of carbon intensity is still 16–20% above the optimum. For
465 operators, clean capacity at a low-carbon site should be treated as a scarce
466 resource with an opportunity cost, much as yield managers protect seats for
467 high-fare late bookings. CARMA sets this protection level implicitly, through
468 ghost jobs learned from the operator’s own history, and needs no hand-tuned
469 reservation thresholds.

470 *Invest in forecasts where they pay.* The value of better intensity forecasts
471 depends on the rest of the policy. For a myopic multi-site planner, perfect
472 forecasts are worth only 1–3 percentage points. Once demand is anticipated,
473 they are worth 4–7 points. When work cannot migrate, they are worth al-
474 most the whole remaining gap. Forecasting effort, for example the integration
475 of numerical weather predictions [33, 32], should therefore follow, not pre-
476 ceede, the fix of demand myopia in multi-site settings. In single-site temporal
477 shifting it is the main lever.

478 *Use risk premia with care.* Calibrated uncertainty is valuable, and ACI kept
479 coverage on target through a year of strong decarbonisation drift. Its best use
480 in scheduling is not obvious, however. A pessimistic carbon price improved
481 a myopic planner by acting as a crude substitute for anticipation, but it
482 degraded CARMA and hurt temporal-only scheduling, where it suppressed
483 good deferrals. Because the planner re-optimises every hour, much of the
484 protection that a risk premium would buy is already provided by recourse.
485 Calibrated quantiles are better suited to reporting and to service-level deci-
486 sions than to inflating costs by default.

487 *Deployment.* CARMA needs three inputs: a public carbon-intensity feed, a
488 demand profile computed from the operator’s own job logs, and a network-
489 flow solver. It decides within tens of milliseconds and returns a plan that
490 is optimal for its model, which makes its decisions auditable. This supports
491 the transparency expected of carbon claims in sustainability reporting. The

492 algorithm is also a rare case in which lower emissions come with better service:
493 under heavy load, CARMA finished far more work on time than the other
494 forecast-driven planners (0.05% late work, against 1.2–2.0%).

495 5.2. Limitations and future research

496 The study has several limitations. First, workloads are synthetic. The
497 generator has realistic features (diurnal and weekly seasonality, heavy-tailed
498 job sizes, a mix of urgent and flexible jobs, width limits), but results on
499 production traces may differ. In particular, CARMA’s advantage depends
500 on how predictable arrivals are, and the sensitivity analysis shows how it
501 degrades when the load deviates from the learned profile. Second, we use
502 average (attributional) regional intensities as published by NESO. Marginal
503 emission factors may be more appropriate for evaluating load shifting [6], and
504 regional values in Great Britain are modelled estimates, not metered data.
505 Third, the facility parameters (PUE, network energy) are stylised, idle server
506 power is not modelled, and embodied emissions are outside the scope of this
507 study [20]. Fourth, the forecaster deliberately excludes weather forecasts, so
508 its day-ahead skill is limited. Fifth, bag-of-tasks jobs with free preemption
509 and migration are an idealisation. Jobs with state, data-gravity or data-
510 sovereignty constraints would add restrictions that break the pure network
511 structure.

512 Several extensions follow naturally. The ghost-demand profile could be
513 replaced by probabilistic arrival forecasts, with the anticipation weight set
514 by a newsvendor-type argument. Scenario-based or distributionally robust
515 variants could treat demand and carbon uncertainty jointly [44]. The frame-
516 work can also be coupled with on-site renewables and storage [19, 20] and
517 with demand-response programmes [21]. Finally, its guarantees could be ex-
518 tended, for instance to competitive bounds for anticipatory policies under
519 bounded forecast error.

520 6. Conclusions

521 This paper studied carbon-aware scheduling of deferrable computing work-
522 loads across data-centre sites as a sustainable-operations problem. We showed
523 that the offline problem, including width limits, deadlines and the emissions
524 of data transfer, is a minimum-cost flow. This yields exact oracles and fast
525 online planning, and we proved that myopic receding-horizon planning can be
526 arbitrarily worse than the oracle. We proposed CARMA, which augments

the planning network with ghost jobs for expected arrivals, and a probabilistic carbon-intensity forecaster with adaptive conformal calibration. In 52 out-of-sample weeks of 2024 on British regional grid data, CARMA cut emissions relative to carbon-agnostic operation by 36–68% when migration was allowed. That is within 6–9% of the clairvoyant optimum and 7.6–9.4% below myopic MPC, and CARMA also reduced late work. Anticipating demand was worth far more than perfect carbon foresight, while risk-averse carbon pricing helped only myopic planners. When jobs cannot migrate, achievable reductions are smaller (9–18%) and forecast accuracy is the binding constraint. Carbon-aware computing is most effective when schedulers plan for the work that has not yet arrived, not only for the grid conditions that lie ahead.

CRediT authorship contribution statement

Quang-Vinh Dang: Conceptualization, Methodology, Software, Data curation, Formal analysis, Investigation, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The carbon-intensity data are publicly available from the NESO Carbon Intensity API [34] under the CC BY 4.0 licence. The data snapshot used in this study, all code, the workload generator with fixed random seeds, and all raw experiment outputs are available from the author’s repository (<https://github.com/vinhqdang/Sustainable-Operations-and-Computers>).

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